In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv("symbols_valid_meta.csv")

df.head()
```

Out[2]:

	Nasdaq Traded	Symbol	Security Name	Listing Exchange	Market Category	ETF	Round Lot Size	Test Issue	Financial Status	Sym
0	Y	A	Agilent Technologies, Inc. Common Stock	N		N	100.0	N	NaN	
1	Y	AA	Alcoa Corporation Common Stock	N		N	100.0	N	NaN	
2	Y	AAAU	Perth Mint Physical Gold ETF	P		Y	100.0	N	NaN	A/
3	Y	AACG	ATA Creativity Global - American Depositary Sh...	Q	G	N	100.0	N	N	I
4	Y	AADR	AdvisorShares Dorsey Wright ADR ETF	P		Y	100.0	N	NaN	A/

In [3]:
```
df.shape
```

Out[3]:
```
(8049, 12)
```

In [4]:
```
df.info
```

Out[4]:
```
<bound method DataFrame.info of      Nasdaq Traded Symbol
Security Name  \
0                 Y      A            Agilent Technologies, Inc. Common Stock
1                 Y     AA                    Alcoa Corporation Common Stock
2                 Y   AAAU                       Perth Mint Physical Gold ETF
3                 Y   AACG  ATA Creativity Global - American Depositary Sh...
4                 Y   AADR                AdvisorShares Dorsey Wright ADR ETF
...             ...    ...                                               ...
8044              Y    ZUO                   Zuora, Inc. Class A Common Stock
8045              Y    ZVO                          Zovio Inc. - Common Stock
8046              Y   ZYME                       Zymeworks Inc. Common Shares
8047              Y   ZYNE       Zynerba Pharmaceuticals, Inc. - Common Stock
8048              Y   ZYXI                          Zynex, Inc. - Common Stock

     Listing Exchange Market Category ETF  Round Lot Size Test Issue  \
0                   N                   N          100.0          N
1                   N                   N          100.0          N
2                   P                   Y          100.0          N
3                   Q               G   N          100.0          N
4                   P                   Y          100.0          N
...               ...             ...  ..            ...        ...
8044                N                   N          100.0          N
8045                Q               Q   N          100.0          N
8046                N                   N          100.0          N
8047                Q               G   N          100.0          N
8048                Q               S   N          100.0          N

     Financial Status CQS Symbol NASDAQ Symbol NextShares
0                 NaN          A             A          N
1                 NaN         AA            AA          N
2                 NaN       AAAU          AAAU          N
3                   N        NaN          AACG          N
4                 NaN       AADR          AADR          N
...               ...        ...           ...        ...
8044              NaN        ZUO           ZUO          N
8045                N        NaN           ZVO          N
8046              NaN       ZYME          ZYME          N
8047                N        NaN          ZYNE          N
8048                N        NaN          ZYXI          N

[8049 rows x 12 columns]>
```

In [5]:
```
df.describe()
```

Out[5]:

	Round Lot Size
count	8049.000000
mean	99.942974
std	2.289194
min	1.000000
25%	100.000000
50%	100.000000
75%	100.000000
max	100.000000

In [11]:
```
df.shape
df.columns
```

Out[11]:
```
Index(['Nasdaq Traded', 'Symbol', 'Security Name', 'Listing Exchange',
       'Market Category', 'ETF', 'Round Lot Size', 'Test Issue',
       'Financial Status', 'CQS Symbol', 'NASDAQ Symbol', 'NextShares'],
      dtype='str')
```

In [12]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 8049 entries, 0 to 8048
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Nasdaq Traded     8049 non-null   str    
 1   Symbol            8049 non-null   str    
 2   Security Name     8049 non-null   str    
 3   Listing Exchange  8049 non-null   str    
 4   Market Category   8049 non-null   str    
 5   ETF               8049 non-null   str    
 6   Round Lot Size    8049 non-null   float64
 7   Test Issue        8049 non-null   str    
 8   Financial Status  3383 non-null   str    
 9   CQS Symbol        4666 non-null   str    
 10  NASDAQ Symbol     8049 non-null   str    
 11  NextShares        8049 non-null   str    
dtypes: float64(1), str(11)
memory usage: 754.7 KB
```

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
Nasdaq Traded          0
Symbol                 0
Security Name          0
Listing Exchange       0
Market Category        0
ETF                    0
Round Lot Size         0
Test Issue             0
Financial Status    4666
CQS Symbol          3383
NASDAQ Symbol          0
NextShares             0
dtype: int64
```

In [16]:
```
import os
os.listdir(r"C:\Users\rktau\finalthiranex")
```

Out[16]:
```
['.ipynb_checkpoints', 'symbols_valid_meta.csv', 'Untitled.ipynb']
```

In [17]:
```
import os
os.listdir(r"C:\Users\rktau\finalthiranex")
```

Out[17]:
```
['.ipynb_checkpoints', 'symbols_valid_meta.csv', 'Untitled.ipynb']
```

In [ ]:
```

```